

COS 433: Cryptography

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Reminders:

Problem Set 0 due today, 11:59pm

Problem Set 1 release today, 11:59pm

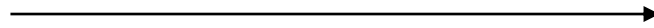
Secret-Key Encryption

Secret-Key Encryption: Syntax



k, m

$$ct = \text{Enc}(k, m)$$



k

$$m' = \text{Dec}(k, ct)$$

Definition: A secret-key encryption scheme (Enc, Dec) consists of two algorithms:

- $\text{Enc}(k, m)$ takes a key $k \in \mathcal{K}$ and message $m \in \mathcal{M}$ and outputs a ciphertext $ct \in \mathcal{C}$
- $\text{Dec}(k, ct)$ takes as input a key and ciphertext and outputs a message m'

A secret-key encryption scheme (Enc, Dec) is **correct** if for all $k \in \mathcal{K}, m \in \mathcal{M}$

$$\text{Dec}(k, \text{Enc}(k, m)) = m$$

Secret-Key Encryption: Security



k, m

$$ct = \text{Enc}(k, m)$$



k

Definition 1: a secret-key encryption scheme (Enc, Dec) is **perfectly secret for uniform messages** if for all functions $\text{Eve}: \mathcal{C} \rightarrow \mathcal{M}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [\text{Eve}(\text{Enc}(k, m)) = m] = \frac{1}{|\mathcal{M}|}$$

Perfect Secrecy

Definition 1': a secret-key encryption scheme (Enc, Dec) is **perfectly secret for uniform messages** if for all messages $m^* \in \mathcal{M}$ and all ciphertexts $ct \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [m = m^* \mid \text{Enc}(k, m) = ct] = \frac{1}{|\mathcal{M}|}$$

This is equivalent to Definition 1. Why?

The best strategy for Eve given ct is to guess the message with the **highest conditional probability** of occurring.

Perfect Secrecy

Definition 1': a secret-key encryption scheme (Enc, Dec) is **perfectly secret for uniform messages** if for all messages $m^* \in \mathcal{M}$ and all ciphertexts $ct \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [m = m^* \mid \text{Enc}(k, m) = ct] = \frac{1}{|\mathcal{M}|}$$

But my messages aren't (uniformly) random!

Perfect Secrecy

Definition 2: a secret-key encryption scheme (Enc, Dec) is **perfectly semantically secure** if for all random variables M over messages, all messages $m^* \in \mathcal{M}$ and all $ct \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m^* \mid \text{Enc}(k, M) = ct] = \Pr_M [M = m^*]$$



k, M

$C = \text{Enc}(k, M)$



k

Perfect Secrecy

Definition 2: a secret-key encryption scheme (Enc, Dec) is **perfectly semantically secure** if for all random variables M over messages, all messages $m^* \in \mathcal{M}$ and all $ct \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m^* \mid \text{Enc}(k, M) = ct] = \Pr_{\substack{M}} [M = m^*]$$

Theorem: a secret-key encryption scheme is perfectly semantically secure if and only if it is perfectly secret for uniform messages!

Perfect Secrecy

Definition 3: a secret-key encryption scheme (Enc, Dec) is **perfectly indistinguishable** if for all message pairs (m_0, m_1)

$$\text{Enc}(k, m_0) \equiv \text{Enc}(k, m_1)$$



k, m_b

$$\text{ct} = \text{Enc}(k, m_b)$$



k

Perfect Secrecy

Definition 3: a secret-key encryption scheme (Enc, Dec) is **perfectly indistinguishable** if for all message pairs (m_0, m_1)

$$\text{Enc}(k, m_0) \equiv \text{Enc}(k, m_1)$$

Theorem: a secret-key encryption scheme is perfectly indistinguishable if and only if it is perfectly secret on uniform messages/perfectly semantically secure.

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Take M to be the uniform distribution

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m \mid \text{Enc}(k, M) = \text{ct}] = \Pr_M [M = m] = \frac{1}{|\mathcal{M}|}$$

Since Eve is only given $\text{Enc}(k, M)$, she has no hope.

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Proof by contradiction. Suppose that $\exists(m_0, m_1)$ such that

$$\text{Enc}(k, m_0) \neq \text{Enc}(k, m_1)$$

Then, $\exists \text{ct}^*$ such that

$$\Pr_{k \leftarrow \mathcal{K}} [\text{Enc}(k, m_0) = \text{ct}^*] > \Pr_{k \leftarrow \mathcal{K}} [\text{Enc}(k, m_1) = \text{ct}^*]$$

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Want to show:
$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [\text{Eve}(\text{Enc}(k, m)) = m] > \frac{1}{|\mathcal{M}|} \quad \text{for some Eve.}$$

Eve(ct):

- If $\text{ct} \neq \text{ct}^*$, guess uniformly from $\mathcal{M}$.
- If $\text{ct} = \text{ct}^*$, guess uniformly from $\mathcal{M}$ but substitute m_0 for m_1
- This is strictly better than uniform guess.

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Want to show:

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m^* \mid \text{Enc}(k, M) = \text{ct}] = \Pr_M [M = m^*]$$

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

Want to show:

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m^* \wedge \text{Enc}(k, M) = \text{ct}] = \Pr_M [M = m^*] \cdot \Pr_{k, M} [\text{Enc}(k, M) = \text{ct}]$$

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$

We know:

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}} [M = m^* \wedge \text{Enc}(k, M) = \text{ct}] = \Pr_{\substack{M \\ k}} [M = m^*] \cdot \Pr_k [\text{Enc}(k, m^*) = \text{ct}]$$

Claim: $\Pr_k [\text{Enc}(k, m^*) = \text{ct}] = \Pr_{k, M} [\text{Enc}(k, M) = \text{ct}]$

Proof: $\forall m, \Pr_k [\text{Enc}(k, m^*) = \text{ct}] = \Pr_k [\text{Enc}(k, m) = \text{ct}]$

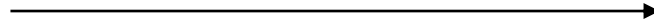
Perfect secrecy seems like a good security notion!

The One-Time Pad



k, m

$$ct = \text{Enc}(k, m)$$



k

$$m' = \text{Dec}(k, ct)$$

Theorem: perfectly secret encryption schemes exist!

Simple proposal: let $\mathcal{K} = \mathcal{M} = \{0,1\}^n$ and define

$$\text{Enc}(k, m) = k \oplus m$$

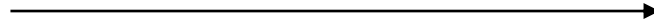
$$\text{Dec}(k, ct) = k \oplus ct$$

The One-Time Pad



k, m

$$ct = \text{Enc}(k, m)$$



k

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Theorem: perfectly secret encryption schemes exist!

Simple proposal: let $\mathcal{K} = \mathcal{M} = \{0,1\}^n$ and define

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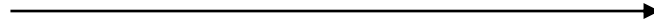
Perfect indistinguishability: $\text{Enc}(\textcolor{blue}{k}, m) \equiv U_n$ (uniform n bit string)

The One-Time Pad



k, m

$$ct = k \oplus m$$



k

$$m' = k \oplus ct$$

Why hasn't everyone been using this scheme all along?

$$\text{Enc}(k, m_1) \oplus \text{Enc}(k, m_2) = m_1 \oplus m_2$$



The One-Time Pad



k, m

$$ct = k \oplus m$$



k

$$m' = k \oplus ct$$

- US and Soviet covert agents used the one-time pad for communication
- Several Soviet spies were discovered by cryptanalysts identifying ciphertexts that re-used one-time pad keys.

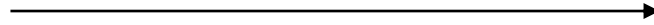
	A	ABCD EFGHIJ KLMNOP QRSTUVWXYZ ZYXWVUTSRQPONMLKJIHGFEDCBA
	B	ABCD EFGHIJ KLMNOP QRSTUVWXYZ YXWVUTSRQPONMLKJIHGFEDCBAZ
	C	ABCD EFGHIJ KLMNOP QRSTUVWXYZ ZSRQPONMLKJIHGFEDCBAZY
	D	ABCD EFGHIJ KLMNOP QRSTUVWXYZ WVUTSRQPONMLKJIHGFEDCBAZT
	E	ABCD EFGHIJ KLMNOP QRSTUVWXYZ VUTSRQPONMLKJIHGFEDCBAZYX
	F	ABCD EFGHIJ KLMNOP QRSTUVWXYZ UTSRQPONMLKJIHGFEDCBAZYXW
	G	ABCD EFGHIJ KLMNOP QRSTUVWXYZ TSRQPONMLKJIHGFEDCBAZYXWV
	H	ABCD EFGHIJ KLMNOP QRSTUVWXYZ SRQPONMLKJIHGFEDCBAZYXWVUT
	I	ABCD EFGHIJ KLMNOP QRSTUVWXYZ RQPONMLKJIHGFEDCBAZYXWVUTS
	J	ABCD EFGHIJ KLMNOP QRSTUVWXYZ QPONMLKJIHGFEDCBAZYXWVUTSR
	K	ABCD EFGHIJ KLMNOP QRSTUVWXYZ PONMLKJIHGFEDCBAZYXWVUTSRQ
	L	ABCD EFGHIJ KLMNOP QRSTUVWXYZ ONMLKJIHGFEDCBAZYXWVUTSRQ
	M	ABCD EFGHIJ KLMNOP QRSTUVWXYZ NMLKJIHGFEDCBAZYXWVUTSRQPO
	N	ABCD EFGHIJ KLMNOP QRSTUVWXYZ MLKJIHGFEDCBAZYXWVUTSRQPON
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	F	ABCD EFGHIJ KLMNOP QRSTUVWXYZ KJIHGFEDCBAZYXWVUTSRQPONML
	Q	ABCD EFGHIJ KLMNOP QRSTUVWXYZ JIHGFEDCBAZYXWVUTSRQPONMLK
	G	ABCD EFGHIJ KLMNOP QRSTUVWXYZ IHGFEDCBAZYXWVUTSRQPONMLKJ
	H	ABCD EFGHIJ KLMNOP QRSTUVWXYZ HGFD CBAZYXWVUTSRQPONMLKJI
	T	ABCD EFGHIJ KLMNOP QRSTUVWXYZ GFEDCBAZYXWVUTSRQPONMLKJIE
	U	ABCD EFGHIJ KLMNOP QRSTUVWXYZ FEDCBAZYXWVUTSRQPONMLKJIEH
	V	ABCD EFGHIJ KLMNOP QRSTUVWXYZ EDCBAZYXWVUTSRQPONMLKJIEHGF
	W	ABCD EFGHIJ KLMNOP QRSTUVWXYZ DCBAZYXWVUTSRQPONMLKJIEHGF
	X	ABCD EFGHIJ KLMNOP QRSTUVWXYZ CBAZYXWVUTSRQPONMLKJIEHGF
	Y	ABCD EFGHIJ KLMNOP QRSTUVWXYZ BAZYXWVUTSRQPONMLKJIEHGFEDC
	Z	ABCD EFGHIJ KLMNOP QRSTUVWXYZ AZYXWVUTSRQPONMLKJIEHGFEDCB
L F N H Y Z A N S E J R N X K S Y M F S K O Z A T		
V R E T H J P C S U R U S Y S J X K N H E L		
P O D T F J J L V J X F E K L N P L S A Z X Y Z		
T S U I O X B N K I N S N D H P N P I O Z V O Z		
E Y J F F O B K K R P N T V Y Y T K S K A T O P R		
N H C J K F P N S V B R Z N Q Q Z Y N C Y S D S		
Y I I U J T W R R Z Q H R D E Y O V R J H O C S Y		
A L O K N H I I N C A I D V R D T E H Z D Z M P		
O I N D S C N O F E X S B V J C A Y S O I S B M U		
K A S Z K O Z J I M D B R C Y B N V Z L F B K T		
T I T I W I F H I N N F R U V V C U I T R N		
N Q O N S Z U B Z B E P V J X N C Z Y F B T E X		
V E I O E H D V T N S S N G L R Z V G U K U G		
P O F R I Q C F A A N L T K E D A N D A G A I N U		
H E I R G L O T E P H V B N X M H U K A C P K A		
A Y G F S Z N F O U S Y R V X I Y I P O R J C E K		
P R O P O J F R I O N Y L I X G T N C Q R X X N		
F S G N A U D T L B U N K A N H A N G T Z Y X N		
U G B O A J X M F Y H T U N H E C T X N O F L S Y		

The One-Time Pad



k, m

$$ct = k \oplus m$$



k

$$m' = k \oplus ct$$

Why hasn't everyone been using this scheme all along?

Corollary: If we want to (eventually) exchange n -bits worth of messages, have to share a key of length n

Can we do better?

- Encrypt *many* messages with perfect secrecy?
- Encrypt long messages with short keys?

No!

Theorem [Shannon]: Any perfectly secret encryption scheme satisfies

$$|\mathcal{K}| \geq |\mathcal{M}|$$

- Applies even if encryption is randomized.
- Extends immediately to “ t -message perfect secrecy:”

$$|\mathcal{K}| \geq |\mathcal{M}|^t$$

Proof of Shannon's Theorem

Intuition: if $|\mathcal{K}| < |\mathcal{M}|$, a uniformly random message “has too much information” to be hidden by a random key.

Proof: let M, M' denote independent, uniformly random messages from $\mathcal{M}$.

Claim:
$$(M, \text{Enc}(K, M)) \equiv (M, \text{Enc}(K, M'))$$

This follows from perfect secrecy.

Proof of Shannon's Theorem

$$(\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{red}{M})) \equiv (\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{green}{M}'))$$

Question: given a pair (m, ct) sampled in this way, what is the probability that there exists a key k such that $\text{Dec}(k, \text{ct}) = m$? ("Plausible pair")

$$\Pr_{\textcolor{blue}{K}, \textcolor{red}{M}} [\exists k: \text{Dec}(k, \text{Enc}(\textcolor{blue}{K}, \textcolor{red}{M})) = \textcolor{red}{M}] = 1 \quad (\text{Correctness})$$

Proof of Shannon's Theorem

$$(\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{red}{M})) \equiv (\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{green}{M}'))$$

Question: given a pair (m, ct) sampled in this way, what is the probability that there exists a key k such that $\text{Dec}(k, \text{ct}) = m$? ("Plausible pair")

$$\begin{aligned} & \Pr_{\textcolor{blue}{K}, \textcolor{red}{M}, \textcolor{green}{M}'} [\exists k: \text{Dec}(k, \text{Enc}(\textcolor{blue}{K}, \textcolor{green}{M}')) = \textcolor{red}{M}] \\ & \leq \sum_k \Pr_{\textcolor{blue}{K}, \textcolor{red}{M}, \textcolor{green}{M}'} [\text{Dec}(k, \text{Enc}(\textcolor{blue}{K}, \textcolor{green}{M}')) = \textcolor{red}{M}] = \frac{|\mathcal{K}|}{|\mathcal{M}|} \end{aligned}$$

Proof of Shannon's Theorem

$$(\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{red}{M})) \equiv (\textcolor{red}{M}, \text{Enc}(\textcolor{blue}{K}, \textcolor{green}{M}'))$$

Question: given a pair (m, ct) sampled in this way, what is the probability that there exists a key k such that $\text{Dec}(k, \text{ct}) = m$? ("Plausible pair")

Conclusion:

$$1 \leq \frac{|\mathcal{K}|}{|\mathcal{M}|}$$

What do we do now?

Perfect Secrecy

Theorem [Shannon]: Any perfectly secret encryption scheme satisfies

$$|\mathcal{K}| \geq |\mathcal{M}|$$

- Applies even if encryption is randomized.
- Extends immediately to “ t -message perfect secrecy:”

$$|\mathcal{K}| \geq |\mathcal{M}|^t$$

Takeaway: perfect secrecy is **too rigid**. How can we weaken the definition?

Statistical Secrecy

Definition: a secret-key encryption scheme (Enc, Dec) is ϵ -statistically secure if for all message pairs (m_0, m_1)

$$\text{Enc}(k, m_0) \approx_{\epsilon} \text{Enc}(k, m_1)$$

For all “distinguishers” $D: \mathcal{C} \rightarrow \{0,1\}$, letting $C_b = \text{Enc}(k, m_b)$,

$$|\mathbf{E}[D(C_0)] - \mathbf{E}[D(C_1)]| \leq \epsilon$$



$\text{ct} = \text{Enc}(k, m_b)$

→



Statistical Secrecy

Theorem [Shannon]: Any statistically secret encryption scheme still requires large keys!

Proof: Problem Set 1

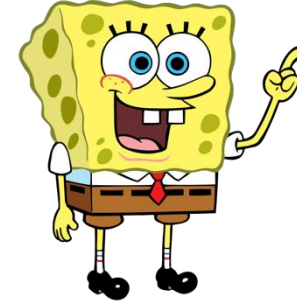
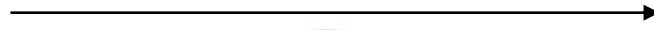
What do we do?

Idea: Assume the Adversary is **Computationally Bounded**

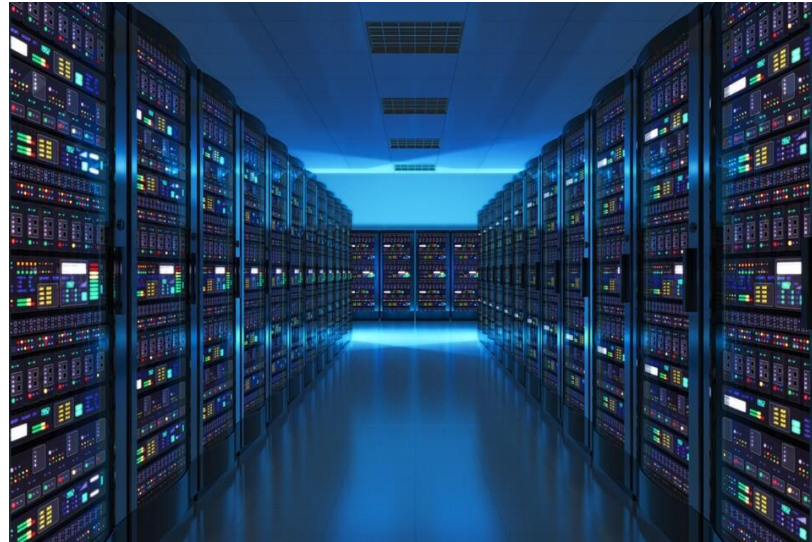


k, m

$$ct = \text{Enc}(k, m)$$



k



(T, ϵ) Indistinguishability

Definition: Two random variables $X \approx_{T, \epsilon} Y$ are (T, ϵ) -computationally indistinguishable if for all “distinguishers” $D: \mathcal{X} \rightarrow \{0,1\}$ computable in time T ,

$$|\mathbf{E}[D(X)] - \mathbf{E}[D(Y)]| \leq \epsilon$$

Definition: a one-time secret-key encryption scheme (Enc, Dec) is (T, ϵ) -computationally secure if for all message pairs (m_0, m_1)

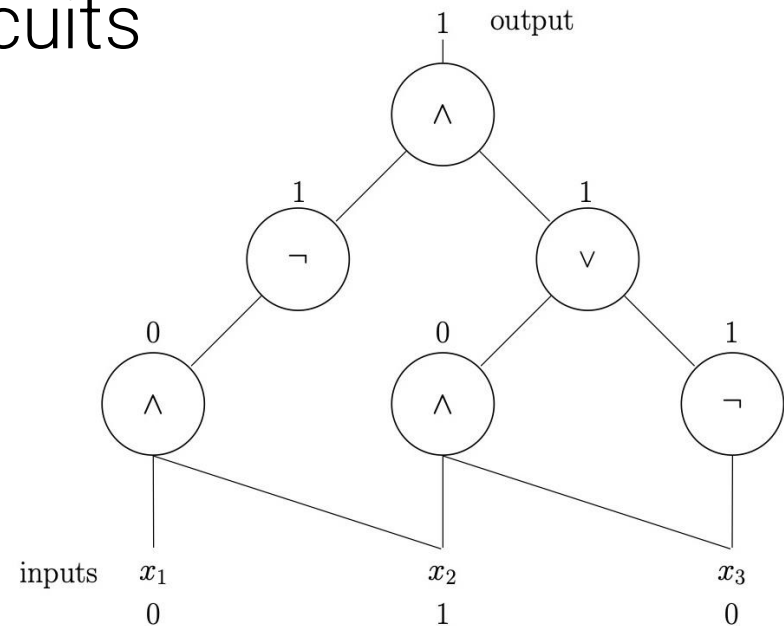
$$\text{Enc}(k, m_0) \approx_{T, \epsilon} \text{Enc}(k, m_1)$$

(T, ϵ) Indistinguishability

What does it mean for D to run in time T ?

Adversary computational model: Boolean circuits

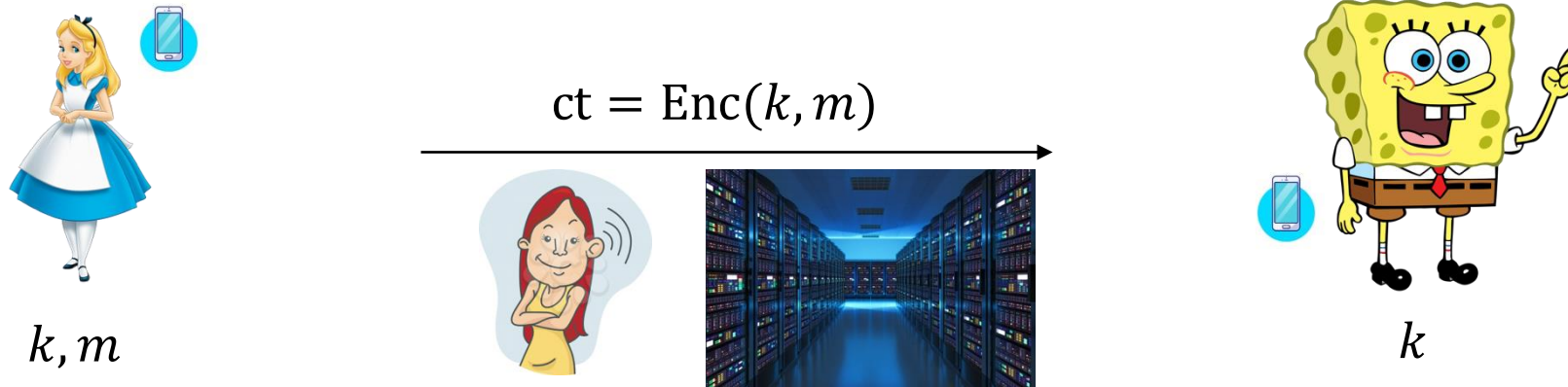
- T = number of gates in the circuit (size)
- Can simulate computation in your favorite programming language (with minimal slowdown).



(T, ϵ) Indistinguishability

General principle:

Assume that the adversary is more powerful than the honest parties.



- Looking for (T, ϵ) -secure encryption schemes where the honest parties run in much less than T time.
- Real-world tradeoffs between efficiency and security.

(T, ϵ) Indistinguishability

Questions:

1. Is the modeling assumption valid? Depends on (T, ϵ) .
2. Do (T, ϵ) -secure encryption schemes exist?
 - The modern world depends on it, but we don't know for sure!
3. How do we build such a scheme?

(T, ϵ) Indistinguishability

Questions:

1. Is the modeling assumption valid? Depends on (T, ϵ) .

Time Estimation (don't try this at home)

Speed of a modern laptop: 2^{37} floating point operations per second.

Speed of a **modern supercomputer**: 2^{60} floating point operations per second.

1000 years = 2^{35} seconds.

Age of the universe: $2^{58.5}$ seconds.

Can execute **2^{120} floating point operations** running a modern supercomputer for the current lifetime of the universe.

In the real world, usually set $T/\epsilon = 2^{128}, 2^{256}, 2^{512}$

Time Estimation (don't try this at home)

Concerns not addressed in this calculation:

- What about parallelization?
- Validity of the Boolean circuit computational model (Church-Turing thesis)
 - Now believed by many to be **false** (e.g., “quantum computers”)
- “Future-proofing”

In the real world, usually set $T/\epsilon = 2^{128}, 2^{256}, 2^{512}$